

Corollary 11.7. *Suppose that $f' = \sum_{\tau} p_{\tau,!}(f^{\tau})$ and $\tilde{f}$ is a nice transfer. Then we have the identity*

$$\frac{1}{2^{1-|S|}L(0, \eta)} \sum_{\tau \in \mathcal{V}_n(E/F)} \sum_{\Pi} J_{\Pi}^{\tau, \tau_0}(f^{\tau}) = \sum_{\pi} I_{\pi}(\tilde{f}),$$

where the left-hand sum ranges over irreducible cuspidal representations Π of $\mathrm{GL}_n(\mathbb{A}_E)$ with central character ω' and the right-hand sum ranges over irreducible cuspidal automorphic representations of $\mathrm{GL}_n(\mathbb{A}_F)$ with central character ω or $\omega\eta$.

To obtain the necessary refined comparison, we make another use of [Ram18, Theorem A]. We introduce the following terminology to simplify a few statements.

Definition 11.8. Fix a character ω of $Z(\mathbb{A}_F)$ and set $\omega' = \omega \circ \mathrm{Nm}_{E/F}$. Fix a split place v_1 and a supercuspidal representation π_{v_1} of $\mathrm{GL}_n(F_{v_1})$. We say that the functions

$$\tilde{f} \in C_c^{\infty}(\mathrm{GL}_n(\mathbb{A}_F) \times \mathbb{A}_{F,n}) \text{ and } \{f^{\tau}\}_{\tau \in \mathcal{V}_n(E/F)} \subset C_c^{\infty}(\mathrm{GL}_n(\mathbb{A}_E))$$

are **efficient transfers** for (ω, π_{v_1}) if they give a ω -nice transfer such that $f_{v_1} \in C_c^{\infty}(\mathrm{GL}_n(F_{v_1}))_{\Omega}$ where Ω is the block of π_{v_1} (f_{v_1} is said to be essentially a matrix coefficient of π_{v_1} in this case) and that $f\tau_{n_{v_1}}$ is related to f_{v_1} as in Proposition 9.12; note that this implies that $f_{v_1}^{\tau}$ is essentially a matrix coefficient of the base change $\Pi_{v_1} \simeq \pi_{v_1} \otimes \pi_{v_1}$ of π_{v_1} .

Making use of the strong multiplicity one theorem [Ram18, Theorem A], we may separate the terms in the preceding sum. For a cuspidal automorphic representation Π of $\mathrm{GL}_n(\mathbb{A}_E)$, we denote by $\mathcal{B}(\Pi)$ the (finite) set of cuspidal automorphic representations π of $\mathrm{GL}_n(\mathbb{A}_F)$ such that $\Pi = \mathrm{BC}_E(\pi)$. We have the following comparison of relative characters.

Proposition 11.9. *For almost all split places v , we fix an irreducible unramified representation π_v^0 . For a fixed split place v_1 , we fix a supercuspidal representation π_{v_1} of $\mathrm{GL}_n(F_{v_1})$. Then there exists at most one cuspidal automorphic representation Π of $\mathrm{GL}_n(\mathbb{A}_E)$ such that if $\tilde{f}$ and $\{f^{\tau}\}_{\tau \in \mathcal{V}_n(E/F)}$ are efficient transfers for (ω, π_{v_1}) , then*

$$\frac{1}{2^{1-|S|}L(0, \eta)} \sum_{\tau \in \mathcal{V}_n(E/F)} J_{\Pi}^{\tau, \tau_0}(f^{\tau}) = \sum_{\pi \in \mathcal{B}(\Pi)} I_{\pi}(\tilde{f}),$$

where the sum on the right runs over all cuspidal automorphic representations π of $\mathrm{GL}_n(\mathbb{A}_F)$ such that

- (1) $\pi_v \cong \pi_v^0$ for almost all split v ,
- (2) π_{v_1} is our fixed supercuspidal representation,

and where $\Pi = \mathrm{BC}_E(\pi)$ is the base change of any π appearing in the sum.

Remark 11.10. Note that the comparison above is vacuous unless $\pi = \bigotimes_v \pi_v$ is GL_n -elliptic at some split place v_2 (this is built into the requirement that $\tilde{f}$ and $\{f^{\tau}\}_{\tau \in \mathcal{V}_n(E/F)}$ are efficient transfers). By Lemma 11.16, it suffices for π_{v_2} to be supercuspidal.